

HANANEEL EPHRAIM MANDAM

US Citizen | BITS Pilani, Hyderabad | github.com/Hananeel-dev | linkedin.com/in/hananeel-ephraim-205b00319 | hananeel.ephraim@gmail.com

SUMMARY

Mechanical engineering student with defense-sector internship experience in real-time physiological signal processing, alongside independent project work in autonomous perception and hands-on powertrain design through Electric Baja SAE. Combines hardware fabrication depth with applied deep learning and computer vision for safety-critical systems.

EDUCATION

B.E. Mechanical Engineering — BITS Pilani, Hyderabad Campus

Expected May 2028

Coursework: Artificial Intelligence, Vibrations & Control, Mechanisms & Machines, Advanced Mechanics of Solids, Fluid Mechanics, Heat Transfer, Differential Equations

EXPERIENCE

Research Intern — Physiological Signal Processing | CSIR-CEERI

May 2026 – Jul 2026

- Built an end-to-end real-time ECG/HRV pipeline integrating commercial health bands to compute KSS drowsiness scores in live rolling windows for a defense project.
- Improved Pan-Tompkins++ R-peak detection achieving 98.8% accuracy vs. 81.7% for NeuroKit2 on the MIT-BIH NSTDB benchmark, a 17-point improvement.
- Designed a pub/sub ECGBus architecture supporting real-time multi-subject BLE session management with hardware-level signal quality gating and automated crash event logging.

Powertrain Engineer & Electrical Lead — Electric Baja SAE | BITS Pilani Hyderabad

Apr 2025 – Apr 2026

- Designed 4:1 inline spur gear reduction and chain-sprocket drivetrain in SolidWorks; validated shaft and gear geometry under 90 Nm peak torque via ANSYS FEA.
- As Electrical Lead, engineered full vehicle safety interlock system with microcontrollers compliant with SAE E-Baja protocols; designed and implemented complete wiring schematic.
- Retrofitted chassis members via profiling and welding for electric drivetrain conversion; coordinated cross-functional team of 7 through fabrication and vendor timeline.

PROJECTS

Uncertainty-Aware Autonomous UAV Emergency Landing Zone Selection

Jun 2026 – Present

- Designed a novel three-channel risk fusion pipeline combining semantic segmentation (SegFormer-B2 fine-tuned on UAVid), epistemic uncertainty via MC Dropout, and temporal label disagreement tracking to select safe UAV landing zones with uncertainty awareness.
- Ablation study across UAVid and Aeroscapes datasets demonstrates that temporal consistency without uncertainty awareness produces false safe zone selections on both datasets; all uncertainty-aware conditions produce zero false safe selections, confirming epistemic uncertainty as the critical safety signal.

3D-Printed Sarrus Linkage | Mechanisms & Machines Coursework

Feb 2026

- Designed, 3D printed, and assembled a functional Sarrus linkage; validated constrained straight-line motion analytically and experimentally.

SKILLS & TOOLS

Programming: Python (PyTorch, NumPy, OpenCV, HuggingFace), MATLAB, Simulink, C, Java

ML & Vision: Semantic Segmentation, Neural Networks, Transfer Learning, Uncertainty Quantification, MC Dropout, Computer Vision

CAD & FEA: SolidWorks, ANSYS (Static Structural, Modal)

Fabrication: MIG/TIG Welding, Lathe, Sheet Metal, FDM 3D Printing, Soldering

Platforms: CUDA / RTX GPU Compute, Git, Linux, Weights & Biases